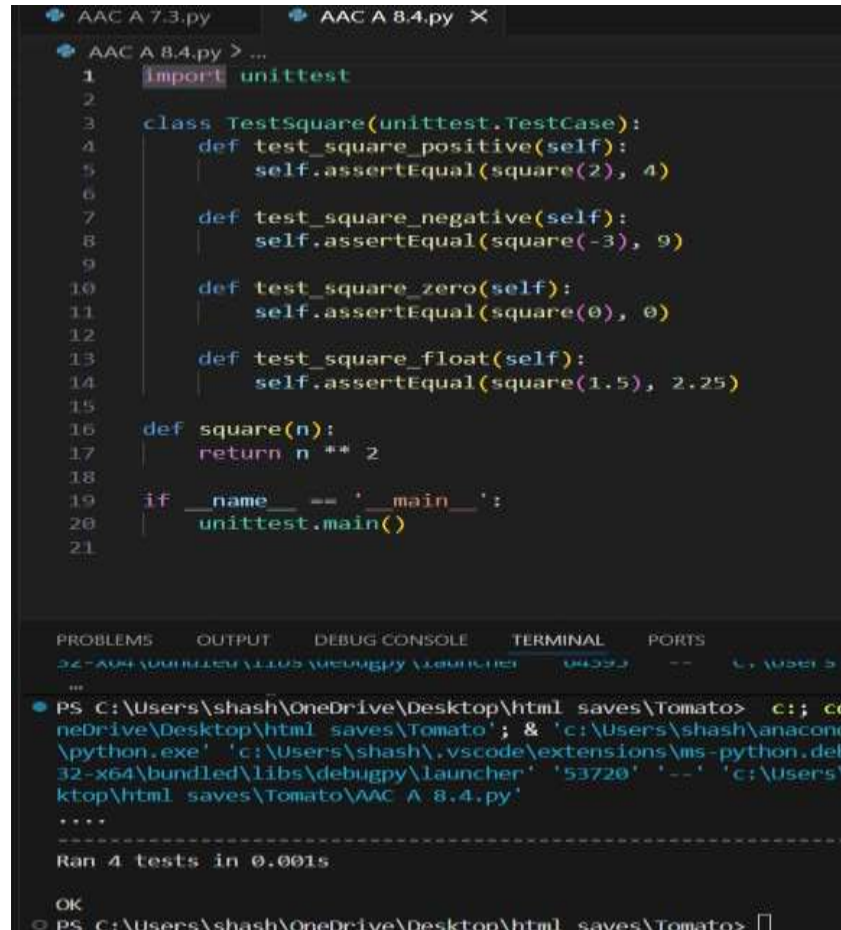


MD.Shehbaaz 2303A51823 Batch-26

SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING																			
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026																		
Course Coordinator Name		Dr. Rishabh Mittal																			
Instructor(s) Name		<table border="1"> <tr><td>Mr. S Naresh Kumar</td></tr> <tr><td>Ms. B. Swathi</td></tr> <tr><td>Dr. Sasanko Shekhar Gantayat</td></tr> <tr><td>Mr. Md Sallauddin</td></tr> <tr><td>Dr. Mathivanan</td></tr> <tr><td>Mr. Y Srikanth</td></tr> <tr><td>Ms. N Shilpa</td></tr> <tr><td>Dr. Rishabh Mittal (Coordinator)</td></tr> <tr><td>Dr. R. Prashant Kumar</td></tr> <tr><td>Mr. Ankushavali MD</td></tr> <tr><td>Mr. B Viswanath</td></tr> <tr><td>Ms. Sujitha Reddy</td></tr> <tr><td>Ms. A. Anitha</td></tr> <tr><td>Ms. M. Madhuri</td></tr> <tr><td>Ms. Katherashala Swetha</td></tr> <tr><td>Ms. Velpula sumalatha</td></tr> <tr><td>Mr. Bingi Raju</td></tr> <tr><td>Mr. G. Kranthi</td></tr> </table>		Mr. S Naresh Kumar	Ms. B. Swathi	Dr. Sasanko Shekhar Gantayat	Mr. Md Sallauddin	Dr. Mathivanan	Mr. Y Srikanth	Ms. N Shilpa	Dr. Rishabh Mittal (Coordinator)	Dr. R. Prashant Kumar	Mr. Ankushavali MD	Mr. B Viswanath	Ms. Sujitha Reddy	Ms. A. Anitha	Ms. M. Madhuri	Ms. Katherashala Swetha	Ms. Velpula sumalatha	Mr. Bingi Raju	Mr. G. Kranthi
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Course Code	23CS002PC304	Course Title	AI Assisted Coding																		
Year/Sem	III/I	Regulation	R23																		
Date and Day of Assignment	Week 4 - Thursday	Time(s)	23CSBTB01 To 23CSBTB52																		
Duration	2 Hours	Applicable to Batches	All Batches																		
Assignment Number: 8.4 (Present assignment number)/24 (Total number of assignments)																					
Q.No.	Question	Expected Time to complete																			
1	Lab 8: Test-Driven Development with AI – Generating and Working with Test Cases	Week 4																			

	Lab Objectives: • To introduce students to test-driven development (TDD) using AI code generation tools. • To enable the generation of test cases before writing code implementations. • To reinforce the importance of testing, validation, and error handling. • To encourage writing clean and reliable code based on AI-generated test expectations. Lab Outcomes (LOs): By the end of this lab, students will be able to: • Apply TDD methodology using AI tools. • Generate test cases before writing the actual code logic. • Validate and refactor code based on test outcomes. • Use Python's unittest or pytest libraries for test-driven development. • Develop confidence in debugging and improving code with AI guidance.	
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	Task 1: Developing a Utility Function Using TDD Scenario You are working on a small utility library for a larger software system. One of the required functions should calculate the square of a given number, and correctness is critical because other modules depend on it. Task Description Following the Test Driven Development (TDD) approach: 1. First, write unit test cases to verify that a function correctly returns the square of a number for multiple inputs. 2. After defining the test cases, use GitHub Copilot or Cursor AI to generate the function implementation so that all tests pass. Ensure that the function is written only after the tests are created. Expected Outcome • A separate test file and implementation file • Clearly written test cases executed before implementation • AI-assisted function implementation that passes all tests • Demonstration of the TDD cycle: <i>test</i> → <i>fail</i> → <i>implement</i> → <i>pass</i>	
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```
1 import unittest
2
3 class TestSquare(unittest.TestCase):
4     def test_square_positive(self):
5         self.assertEqual(square(2), 4)
6
7     def test_square_negative(self):
8         self.assertEqual(square(-3), 9)
9
10    def test_square_zero(self):
11        self.assertEqual(square(0), 0)
12
13    def test_square_float(self):
14        self.assertEqual(square(1.5), 2.25)
15
16    def square(n):
17        return n ** 2
18
19    if __name__ == '__main__':
20        unittest.main()
21
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash\OneDrive\Desktop\html_saves\Tomato> c:\Users\shash\OneDrive\Desktop\html_saves\Tomato> python AAC A 8.4.py

....

Ran 4 tests in 0.001s

OK

PS C:\Users\shash\OneDrive\Desktop\html_saves\Tomato>

Task 2: Email Validation for a User Registration System

Scenario

You are developing the backend of a user registration system. One requirement is to validate user email addresses before storing them in the database.

Task Description

Apply **Test Driven Development** by:

1. Writing unit test cases that define valid and invalid email formats (e.g., missing @, missing domain, incorrect structure).
2. Using **AI assistance** to implement the validate_email() function based strictly on the behavior described by the test cases.

The implementation should be driven entirely by the test expectations.

Expected Outcome

- Well-defined unit tests using unittest or pytest
- An AI-generated email validation function
- All test cases passing successfully
- Clear alignment between test cases and function behavior

```
AAC A 7.3.py AAC A 8.4.py X
AAC A 8.4.py > ...
1 import unittest
2 import re
3
4 class TestEmailValidation(unittest.TestCase):
5     def test_valid_email(self):
6         self.assertTrue(validate_email("user@example.com"))
7
8     def test_email_without_at(self):
9         self.assertFalse(validate_email("userexample.com"))
10
11    def test_email_without_domain(self):
12        self.assertFalse(validate_email("user@"))
13
14    def test_email_with_invalid_chars(self):
15        self.assertFalse(validate_email("user@exam ple.com"))
16
17    def test_email_with_multiple_at(self):
18        self.assertFalse(validate_email("user@example.com"))
19
20    def validate_email(email):
21        pattern = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
22        return re.match(pattern, email) is not None
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AAC A 7.3.py AAC A 8.4.py X
AAC A 8.4.py > TestMaxOfThree > test_max_first_max
1 import unittest
2
3 class TestMaxOfThree(unittest.TestCase):
4     def test_max_all_positive(self):
5         self.assertEqual(max_of_three(1, 2, 3), 3)
6
7     def test_max_with_negatives(self):
8         self.assertEqual(max_of_three(-1, -2, -3), -1)
9
10    def test_max_mixed(self):
11        self.assertEqual(max_of_three(-1, 5, 0), 5)
12
13    def test_max_duplicates(self):
14        self.assertEqual(max_of_three(2, 2, 2), 2)
15
16    def test_max_first_max(self):
17        self.assertEqual(max_of_three(10, 5, 7), 10)
18
19    def max_of_three(a, b, c):
20        return max(a, b, c)
21
22    if __name__ == '__main__':
23        unittest.main()
24
25
26 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
-----
Ran 5 tests in 0.001s
OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
-----
Ran 5 tests in 0.000s
OK
```

Task 4: Shopping Cart Development with AI-Assisted TDD

Scenario

You are building a simple shopping cart module for an e-commerce application. The cart must support adding items, removing items, and calculating the total price accurately.

Task Description

Follow a **test-driven approach**:

1. Write unit tests for each required behavior:
 - Adding an item
 - Removing an item
 - Calculating the total price
2. After defining all tests, use **AI tools** to generate the ShoppingCart class and its methods so that the tests pass.

Focus on behavior-driven testing rather than implementation details.

Expected Outcome

- Unit tests defining expected shopping cart behavior
- AI-generated class implementation
- All tests passing successfully
- Clear demonstration of TDD applied to a class-based design

```

AAC A 7.3.py AAC A 8.4.py X
AAC A 8.4.py > TestShoppingCart > test_calculate_total_empty
1 import unittest
2
3 class TestShoppingCart(unittest.TestCase):
4     def setUp(self):
5         self.cart = ShoppingCart()
6
7     def test_add_item(self):
8         self.cart.add_item(("apple", 1.0))
9         self.assertEqual(len(self.cart.items), 1)
10        self.assertEqual(self.cart.items[0], ("apple", 1.0))
11
12    def test_remove_item(self):
13        self.cart.add_item(("apple", 1.0))
14        self.cart.remove_item(("apple", 1.0))
15        self.assertEqual(len(self.cart.items), 0)
16
17    def test_calculate_total_empty(self):
18        self.assertEqual(self.cart.calculate_total(), 0.0)
19
20    def test_calculate_total_with_items(self):
21        self.cart.add_item(("apple", 1.0))
22        self.cart.add_item(("banana", 0.5))
23        self.assertEqual(self.cart.calculate_total(), 1.5)
24
25
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
-----
Ran 5 tests in 0.000s
OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
....
-----
Ran 4 tests in 0.000s
OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato>

```



```
AAC A 7.3.py AAC A 8.4.py X
AAC A 8.4.py > TestShoppingCart > test_calculate_total_empty
1 class TestShoppingCart(unittest.TestCase):
28 def test_calculate_total_with_items(self):
21     self.cart.add_item(("apple", 1.0))
22     self.cart.add_item(("banana", 0.5))
23     self.assertEqual(self.cart.calculate_total(), 1.5)
24
25 class ShoppingCart:
26     def __init__(self):
27         self.items = []
28
29     def add_item(self, item):
30         self.items.append(item)
31
32     def remove_item(self, item):
33         if item in self.items:
34             self.items.remove(item)
35
36     def calculate_total(self):
37         return sum(price for _, price in self.items)
38
39 if __name__ == '__main__':
40     unittest.main()
41
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
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Ran 5 tests in 0.000s
OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
....
Ran 4 tests in 0.000s
OK
```

Task 5: String Validation Module Using TDD

Scenario

You are working on a text-processing module where a function is required to identify whether a given string is a palindrome. The function must handle different cases and inputs reliably.

Task Description

Using **Test Driven Development**:

1. Write test cases for a palindrome checker covering:
 - Simple palindromes
 - Non-palindromes
 - Case variations
2. Use **GitHub Copilot or Cursor AI** to generate the `is_palindrome()` function based on the test case expectations.

The function should be implemented only after tests are written.

Expected Outcome

- Clearly written test cases defining expected behavior

- AI-assisted implementation of the palindrome checker
- All test cases passing successfully
- Evidence of TDD methodology applied correctly

```

AAC A 7.3.py AAC A 8.4.py
AAC A 8.4.py > ...
1 import unittest
2 class TestPalindrome(unittest.TestCase):
3     def test_simple_palindrome(self):
4         self.assertTrue(is_palindrome("radar"))
5
6     def test_non_palindrome(self):
7         self.assertFalse(is_palindrome("hello"))
8
9     def test_case_insensitive(self):
10        self.assertTrue(is_palindrome("Racecar"))
11
12    def test_empty_string(self):
13        self.assertTrue(is_palindrome(""))
14
15    def test_single_character(self):
16        self.assertTrue(is_palindrome("a"))
17
18    def is_palindrome(s):
19        s = s.lower()
20        return s == s[::-1]
21
22    if __name__ == '__main__':
23        unittest.main()
24
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
-----
Ran 4 tests in 0.000s

OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato> python "AAC A 8.4.py"
.....
-----
Ran 5 tests in 0.000s

OK
PS C:\Users\shash\OneDrive\Desktop\html saves\Tomato>

```

Note: Report should be submitted a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots